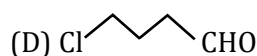
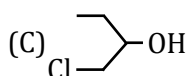
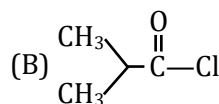
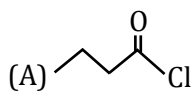




## Organic Compounds Containing Nitrogen : RACE-01

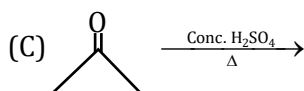
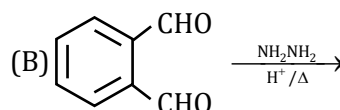
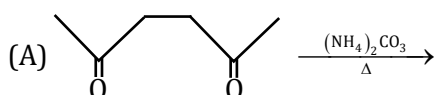
## Reaction Involving Preparation of Organic Compound Containing Nitrogen

1.  $(X)C_4H_7OCl \xrightarrow{NH_3} C_4H_9ON \xrightarrow[\text{KOH}]{Br_2} CH_3CH_2CH_2NH_2$ ; Compound (X) is :



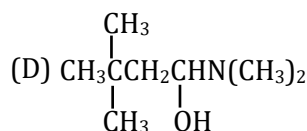
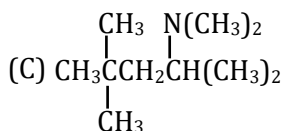
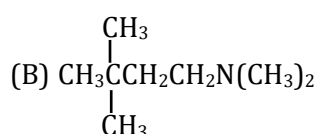
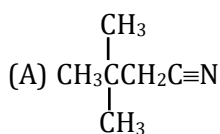
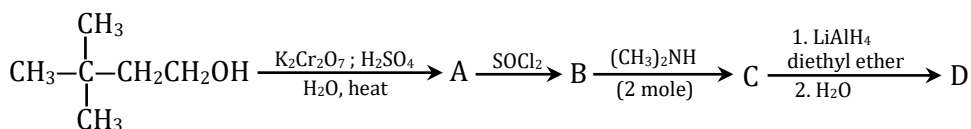
CCN150

2. Identify reactions that give aromatic products.



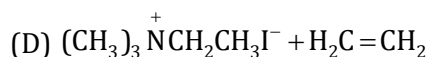
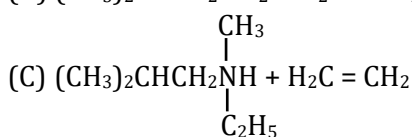
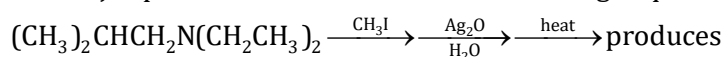
CCN151

3. Identify product D in the following reaction sequence :



CCN153

4. The major products obtained from the following sequence of reactions are:

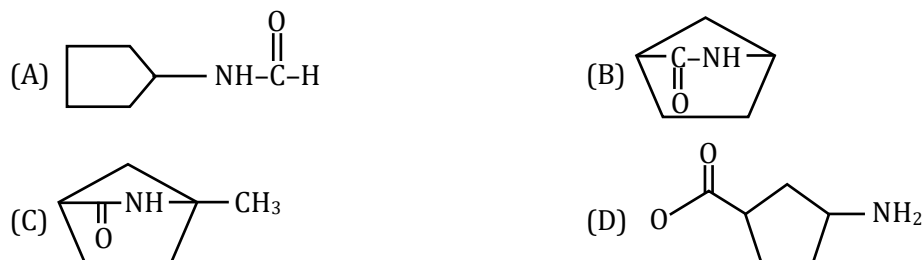


CCN154

5. A nitrile X is treated with  $\text{LiAlH}_4$  to obtain compound Y ( $\text{C}_2\text{H}_7\text{N}$ ). In a separate reaction X is hydrolyzed in an acid medium to obtain Z. The product obtained after mixing Y and Z will be?
- (A)  $\text{CH}_3\text{CONHCH}_2\text{CH}_3$  (B)  $\text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3$   
 (C)  $(\text{CH}_3\text{COO}^-)(\text{CH}_3\text{CH}_2\text{NH}_3^+)$  (D)  $(\text{CH}_3\text{CH}_2\text{COO}^-)(\text{CH}_3\text{NH}_2^+)$

CCN156

6. Compound (B) is :



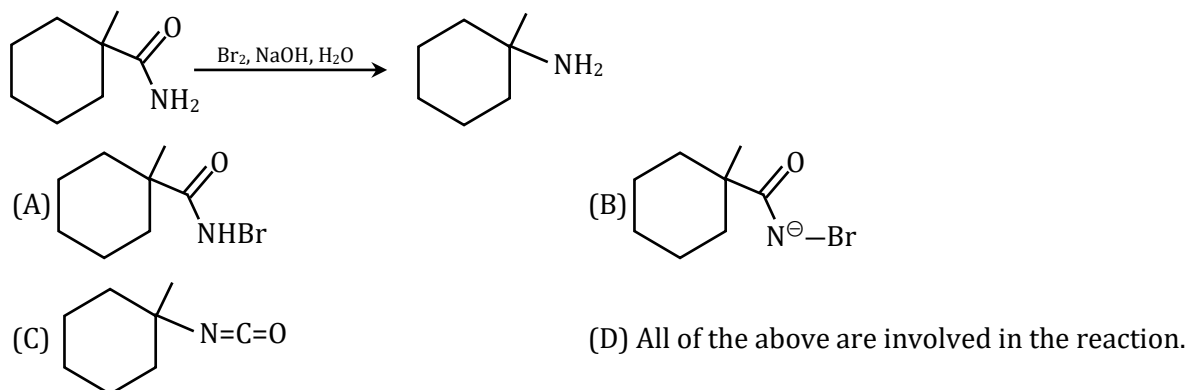
CCN158

7. Which of the following will not give Hoffmann bromamide reaction ?



CCN159

8. Which of the following species would not be involved in the Hoffmann rearrangement shown below ?



CCN160

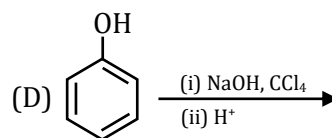
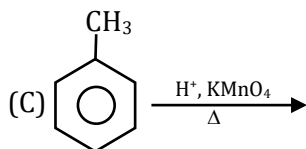
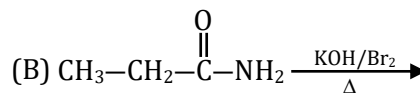
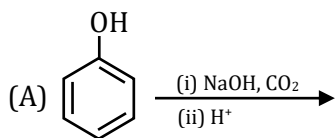
9.  $\text{CH}_3\text{CH}_2\text{CN} \xrightarrow[\text{H}_2\text{O}]{\text{LiAlH}_4} \text{X}$   
 $\text{CH}_3\text{CH}_2\text{NC} \xrightarrow[\text{H}_2\text{O}]{\text{LiAlH}_4} \text{Y}$

Identify the relation between product X & Y.

- (A) Stereoisomers (B) Functional group isomers  
 (C) Positional isomers (D) Chain isomers

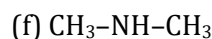
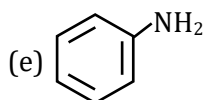
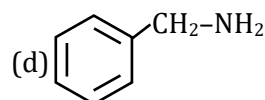
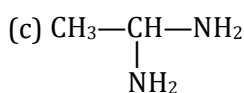
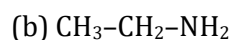
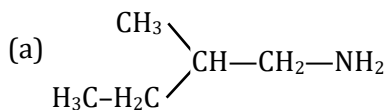
CCN191

10. Which of the following has  $-\text{COOH}$  group in major product ?



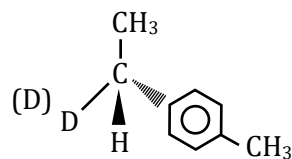
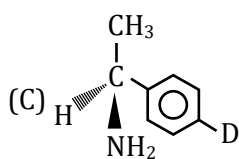
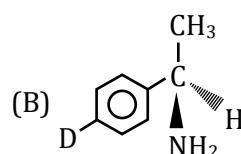
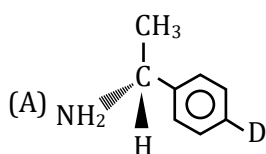
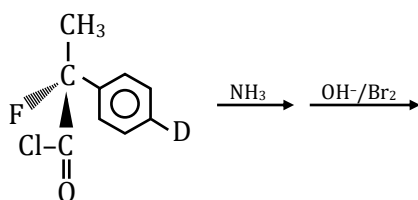
CCN192

11. The total number of amines among following which can be synthesized by Gabriel synthesis



CCN193

12.



CCN194

## ANSWER KEY

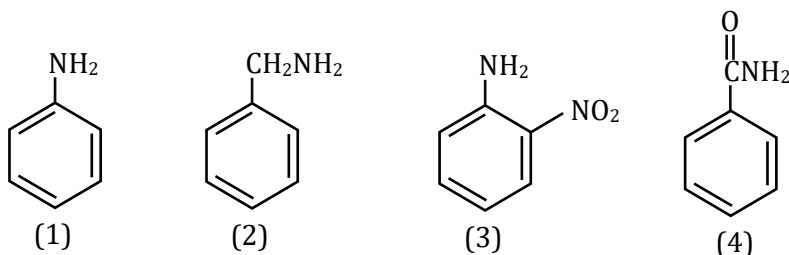
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|----------|----|-----|---|---|---|---|---|---|---|-----|
| Answer   | A  | ABC | B | C | C | B | D | D | B | ACD |
| Question | 11 | 12  |   |   |   |   |   |   |   |     |
| Answer   | 4  | C   |   |   |   |   |   |   |   |     |



Organic Compounds Containing Nitrogen : RACE -02

Reaction Involving Organic Compound Containing Nitrogen

1. Rank the following compounds in order of increasing basic strength. (weakest  $\rightarrow$  strongest):



- (A)  $4 < 2 < 1 < 3$       (B)  $4 < 3 < 1 < 2$       (C)  $4 < 1 < 3 < 2$       (D)  $2 < 1 < 3 < 4$

CCN152

2.  $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow{\text{NH}_3} \xrightarrow{\Delta} \xrightarrow{\text{Br}_2/\text{KOH}} \text{(A)}$

Which is not correct statement for (A)

- (A) It gives carbyl amine reaction.      (B) It gives mustard oil reaction.  
(C) Forms alcohol with  $\text{NaNO}_2/\text{HCl}$       (D) Forms yellow oily liquid with  $\text{NaNO}_2/\text{HCl}$

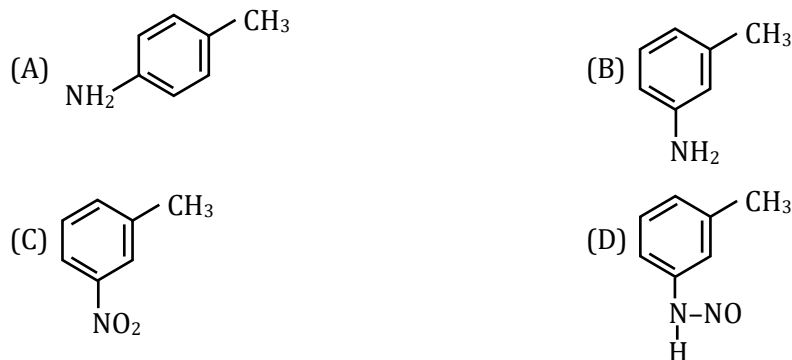
CCN195

3. Which of the following is most reactive ion in azocoupling reaction ?



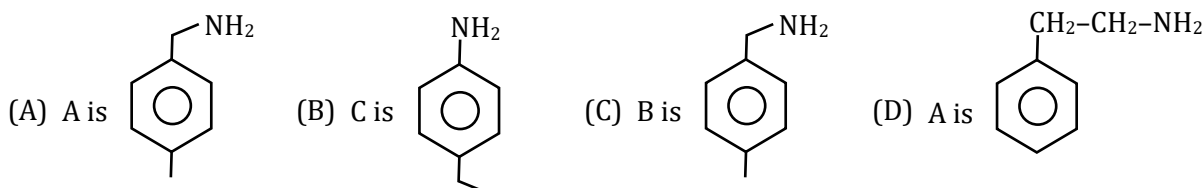
CCN196

4.  $\xrightarrow[2. \text{H}_3\text{PO}_2/\text{H}_2\text{O}]{1. \text{NaNO}_2+\text{HCl}}$  A ; 'A' is



CCN197

5. Three bottles A, B and C have been found, each of which contains a liquid and is labeled "amine  $C_8H_{11}N$ ." As an expert in amine chemistry, you have been hired as a consultant and asked to identify each compound. Compounds A and B give off a gas when they react with  $NaNO_2$  and  $HCl$  at  $0^\circ C$ ; C does not. However, when the aqueous reaction mixture from the diazotization of C is warmed, a gas is evolved. Compound A is optically inactive. Oxidation of C with  $H_2O_2$  (a reagent known to oxidize amino groups to nitro groups), followed by vigorous oxidation with  $KMnO_4$  gives p-nitrobenzoic acid. Oxidation of B in a similar manner yields 1,4-benzene-dicarboxylic acid (terephthalic acid), and oxidation of A yields benzoic acid. Identify compounds A, B and C.

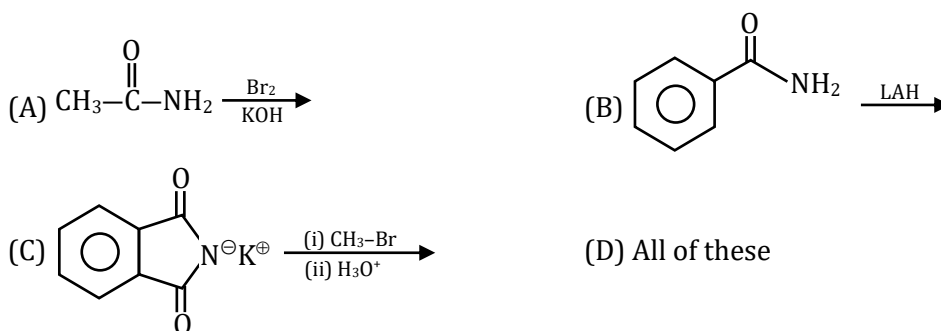


CCN198

6. What is the number of benzene derivatives having the molecular formula  $C_7H_9N$  that would respond positive to carbylamine test.

CCN199

7. In which of the following reaction, one of the product obtained will give Hoffmann mustard oil test?

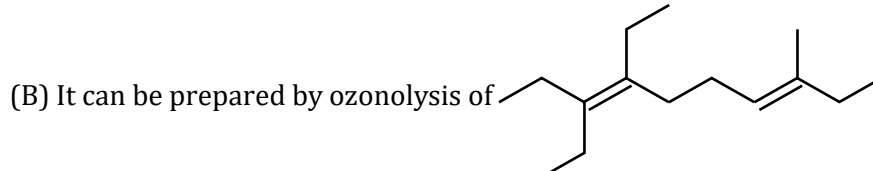


CCN201

8.  $H_3C-\underset{\substack{| \\ OH}}{CH}-\underset{\substack{| \\ NH_2}}{CH}-CH_3 \xrightarrow{HNO_2} J$  (Major product)

Which statement is incorrect about above major product ?

- (A) It can be prepared by reaction of But-2-yne with  $Hg^{2+} / H^+ / H_2O$

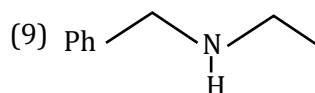
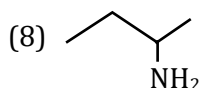
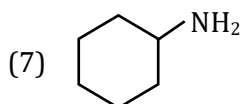
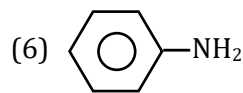
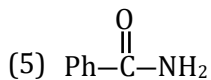
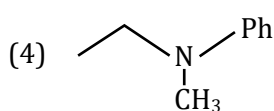
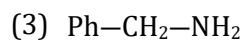
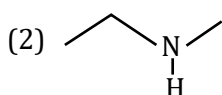
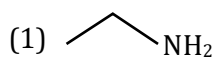


- (C) J on reaction with  $Mg-Hg$  followed by acidification gives  $Me-\underset{\substack{| \\ OH}}{\overset{\substack{Me \\ |}}{C}}-\underset{\substack{| \\ OH}}{\overset{\substack{Et \\ |}}{C}}-Et$  as one of the product.

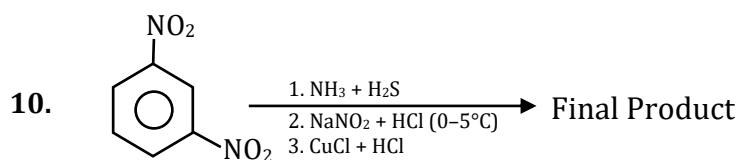
- (D) It can form an oxime on reaction with  $NH_2-OH$  which shows geometrical isomerism.

CCN202

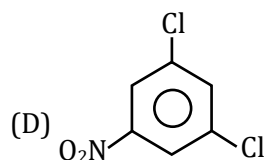
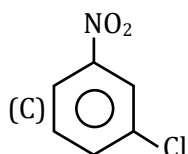
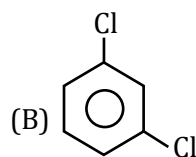
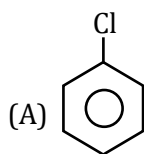
9. Which of the following will show positive carbyl amine test ?



CCN203

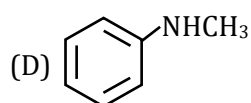
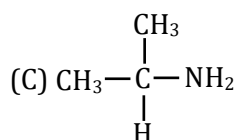
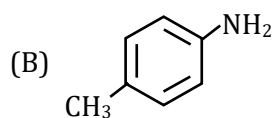
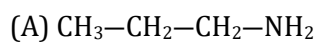


Final Product is?



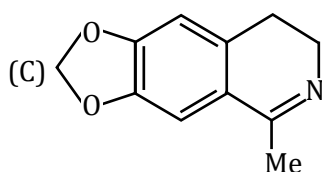
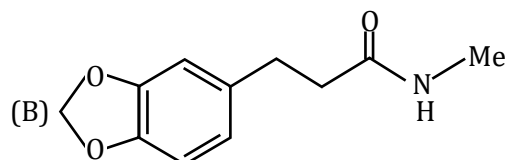
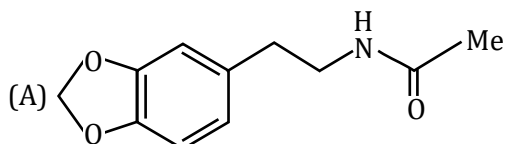
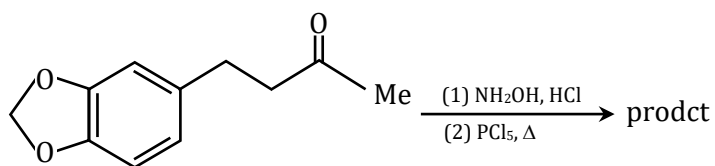
CCN204

11. Which one of the following gives the most stable Diazonium salt ?



CCN205

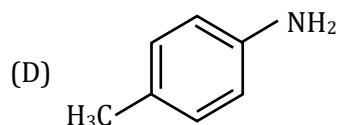
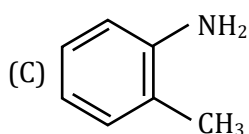
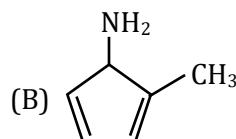
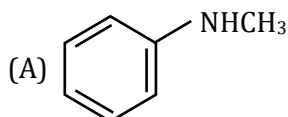
12. Which of the following product will be obtained in the given (consider minor product also) Beckmann-type rearrangement?



(D) All of these

CCN155

13. The compound X ( $\text{C}_7\text{H}_9\text{N}$ ) reacts with benzenesulfonyl chloride to give Y ( $\text{C}_{13}\text{H}_{13}\text{NO}_2\text{S}$ ) which is insoluble in alkali. The compound X is -



CCN157

## ANSWER KEY

| Question | 1  | 2  | 3  | 4 | 5   | 6 | 7 | 8 | 9 | 10 |
|----------|----|----|----|---|-----|---|---|---|---|----|
| Answer   | B  | D  | D  | C | BCD | 4 | D | C | 5 | C  |
| Question | 11 | 12 | 13 |   |     |   |   |   |   |    |
| Answer   | B  | D  | A  |   |     |   |   |   |   |    |